INCH-POUND

MIL-P-70732 (AR)

31 MARCH 1989

MILITARY SPECIFICATION

PROPELLANT, STICK, M31A1E1
FOR USE IN CHARGE, PROOF, 155MM, PXR-6297A1

This Specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers M31A1E1 stick propellant for use in the 155MM propelling charge designated as charge, PXR6297A1 for the 155MM Cannon, M199/Howitzer, M198.

1.2 Classification. The propellant shall be Type II, cylindrical, slotted, single-perforated stick.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1376

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MIL-P-70456 - Propellant, Stick, M31A1E1 For Use In Charge Propelling M203A1
MIL-C-70729 - Charge Proof, 155MM, PXR-6297A1 Loading, Assembling and Packing

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12910074 - Propellant and Liner Assembly

(Copies of specifications, standards, drawings, publications and other Government documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. The propellant shall be M31A1E1, conforming to the requirements of MIL-P-70456 and Dwg 12910074.

3.2 Proving ground.

3.2.1 Ballistic assessment requirements. The propellant, when fired in the M185 or M199 cannon (with 95 percent minimum life remaining tube) and with the projectile (M549 and M864) (See 6.4) and charges manufactured in accordance with MIL-C-70729 shall comply with the following requirements when tested as specified in 4.5.1.1. At 70°F and standard weapon conditions, the propellant shall be capable of assessment to an average pressure of 58,800 pounds per square inch (psi) for the M549 projectile. The test which produces a M549 projectile pressure

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assessment of 58,800 psi shall produce a corresponding average pressure (60,300 psi informational) when tested with the M864 projectile.

3.2.2 Ballistic uniformity requirements. The propellant, when fired in the M199 cannon (with 95 percent minimum life remaining tube) at the established charge weight and with the projectile (M549 or M864) (see 6.4) for which it was manufactured, shall comply with the requirements (see 3.2.1) when tested as specified in 4.5.1.2.

3.3 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitbale for the performance of the inspection requirements specified herein, unless diapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absense of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. Quality conformance inspection shall be conducted on this item.

4.3 Quality conformance inspection.

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4.3.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078.

4.3.2 Examination and testing.

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.3.2.1 Sampling for laboratory testing (see 3.1). Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical/physical properties in accordance with MIL-P-70456.

4.3.2.2 Sampling for ballistic testing (see 3.2). Each lot of propellant shall be sampled for Proving Ground tests in accordance with MIL-P-70456.

4.3.2.3 Charge establishment (see 3.2). If the lot does not meet the applicable requirements, testing shall cease and the Test Director shall immediately notify the procuring activity.

4.3.2.4 Average pressure (see 3.2.2). If any average pressure obtained at 70°F exceeds the applicable requirement, apply retest criteria in accordance with 4.5.1.3.

4.4 Inspection equipment. The provisions of MIL-A-48078 shall apply.

4.5 Methods of inspection.

4.5.1 Proving ground assessments (see 6.5). These tests shall be conducted at a Government proving ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedures. The following tolerances shall apply to 70°F conditioning: 70°F from 68°F to 72°F.

4.5.1.1 Charge establishment (See Figure 1). For each of the projectiles, M549 and M864, three or more groups of rounds loaded with selected charge weights of propellant (weights selected should bracket expected established charge weight) shall be conditioned and fired at 70°F. From the data obtained, a charge weight shall be calculated which will yield the prescribed pressure (see 3.2). This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.5.1.2). The difference between the corrected pressure obtained during the uniformity series and the service pressure shall be compensated for by a final adjustment in charge weight using the slope of the charge weight curve previously established. The resultant will be the recommended charge weight.

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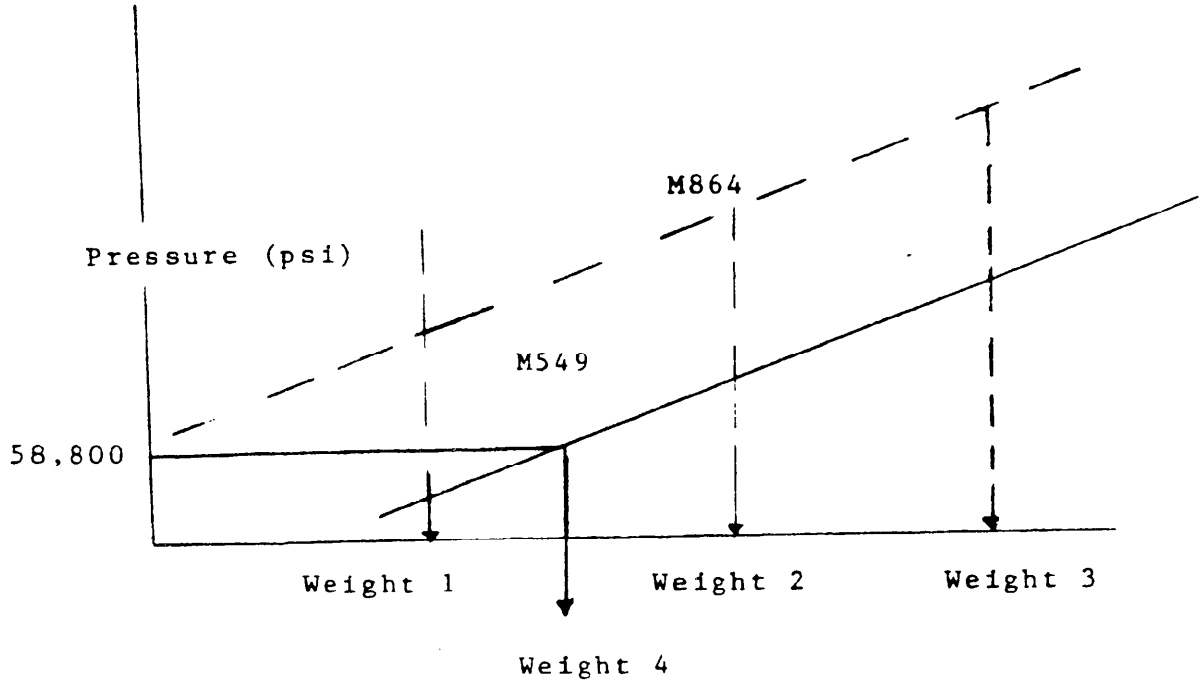


Figure 1 Test Matrix

During charge establishment 6 charges at each weight 1, 2, and 3 shall be loaded. Three charges at each weight shall be fired with the M549, and the three with the M864 to determine the charge slopes. The slope of the M549 pressure weight curve shall be used to determine the propellant weight (weight 4) at 58,800 psi pressure. The corresponding pressure on the M864 slope shall be determined after the M549 ten round uniformity test.

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4.5.1.2 Uniformity (See Figure 1). For the M549 projectile, ten rounds shall be assembled with the established charge weight of propellant and shall be conditioned to 70°F (as indicated by thermocoupled control round). Firing results shall be corrected to standard projectile weight (see 3.2), to recommended charge weight conditions (i.e., the difference between as fired charge weight and recommended charge weight) and for deviations of the pressure shall be calculated using corrected results.

4.5.1.3 Retest criteria (see Figure 1). If for any reason the proving ground considers that test conditions have detrimentally affected the test results, additional charges as required shall be tested. A lot shall be retested whenever any of the following conditions exist and whenever the proving ground deems it necessary (see 6.3).

a. Test lot fails average pressure at 70°F.

b. The final adjustment in charge weight exceeds two percent of the charge weight used in uniformity firings, retest for charge establishment in accordance with 4.5.1.1.

5. PACKAGING

5.1 Packing.

5.1.1 Level A - The propellant shall be packed in accordance with MIL-P-70456.

5.1.2 Level B - The propellant shall be packed in accordance with MIL-P-70456.

5.1.3 Level C - The propellant shall be packed in accordance with MIL-P-70456.

5.2 Marking.

5.2.1 Levels A, B, and C. - Mark in accordance with MIL-P-70456.

6. NOTES

6.1 Intended Use. The proof charge covered by this specification is intended for use with 155MM projectile development and acceptance tests and for M199 Cannon Proof Tests. This charge is intended for Proving Ground use only to be fired in a M199 Cannon with all personnel behind a protective barrier.

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6.2 Data requirements. One copy of the Propellant Description Sheet and one copy of all ballistic test data for each lot of propellant shall be forwarded to ARDEC, Picatinny Arsenal, N.J. 07806-5000, ATTN: AMSMC-QAR-R(D), AMCPM-CAWS-A and SMCAR-AEE-BP.

6.3 Test anomalies. Any unusual ocurrences during ballistic testing, such as excessive delays between rounds, stickers, poor projectile flights, etc., shall be reported in detail on the applicable firing record.

6.4 Projectile weight. All projectiles intended for use in test rounds shall be inert-loaded to a tolerance of ± .01 lb. (4.5 gm). The Loading Authorization for the propellant lot should be annotated to indicate that the service charge weight listed is to be used only when loading with projectiles having the listed weight. If the projectile weight is different from that listed, the lot should be reassessed to the new projectile weight or the charge weight adjusted using a known projectile weight/charge weight curve.

6.5 Proving ground test summary.

TABLE I

PROVING GROUND TEST SUMMARY

Temperature (°F)	Sample Size	Average Vel (fps)	Std Dev of Press (psi)	Average Pressure (psi)	Differential Pressure
			CHARGE ESTABLISHMENT		
70	9 (M549)	Info.	N/A	58,800	Info.
	9 (M864)	Info.	N/A	60,300 (Informational)	Info.
			UNIFORMITY		
70	10 (M549)	Info.	Info.	58,800	Info.

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6.6 Subject term (Key word) listing.

Artillery
Stick propellant
Proof Charge
Howitzer
Projectile
Cannon

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1376-A274)

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-70732(AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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